

Part1

Answers:

TBD (Can't find instructions for edge impulse)

Part 2

Discussion questions

What threshold did you choose for anomaly detection, and why?

2.0, because it is larger than the normal max but lower than the anomaly min and much lower than the anomaly mean.

- How does reconstruction error help distinguish normal and anomalous activity?

The autoencoder is only trained on the normal data which means it reconstructs normal windows with high accuracy but fails on the anomaly patterns it sees.

- What information from the Python notebook must be preserved when deploying the model to Arduino?

The in8t model weights need to be preserved along with the quantization scale and zero point and the threshold which I selected to be 2.0.

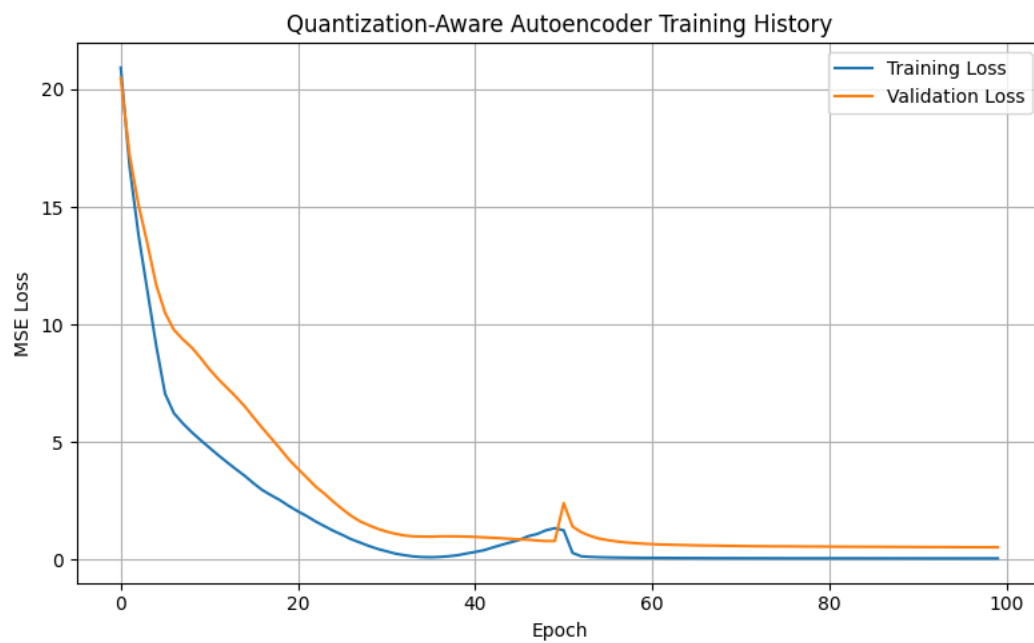
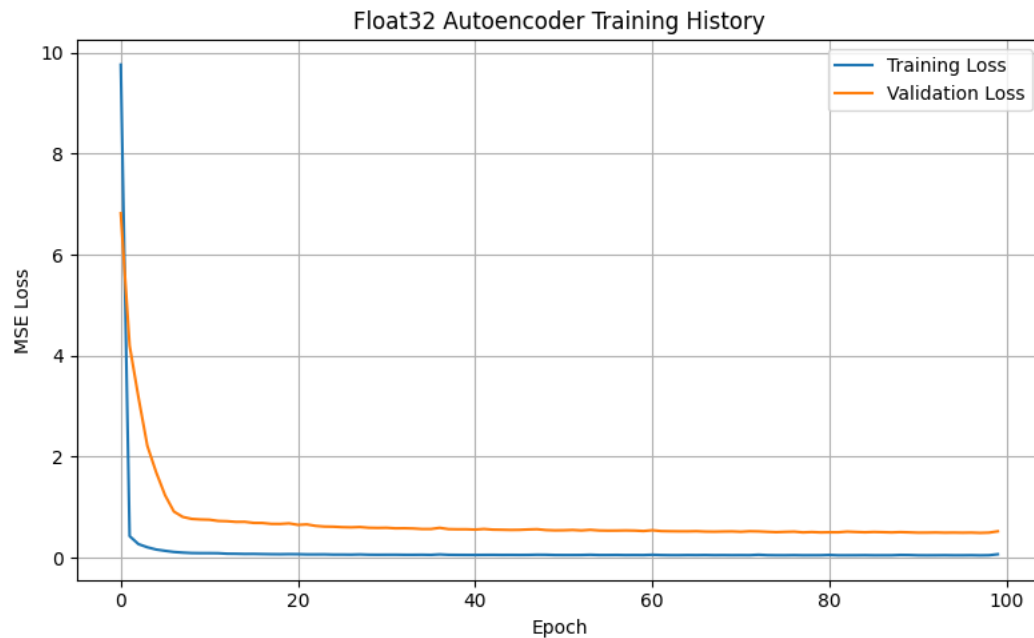
- Why is it important for the Arduino sketch to use the same preprocessing assumptions as the notebook?

Because the model was trained on the 100 sample data windows which it used to learn the normalization, window size and zeros. If there is deviation from these values then the threshold value is meaningless and arbitrary.

- What are the main limitations of this anomaly detection method when deployed on a microcontroller?

The threshold is fixed so changes in environment, user or other factors could push the values in either direction making the threshold less accurate.

Notebook outputs:



Train normal	mean=0.100061, stderr=0.001684, min=0.008211, max=1.148123
Train anomaly	mean=42.047409, stderr=0.372207, min=2.471123, max=134.862518
Test normal	mean=0.137435, stderr=0.003920, min=0.008787, max=1.356646
Test anomaly	mean=42.454029, stderr=0.623866, min=2.767509, max=133.250809



Train set evaluation

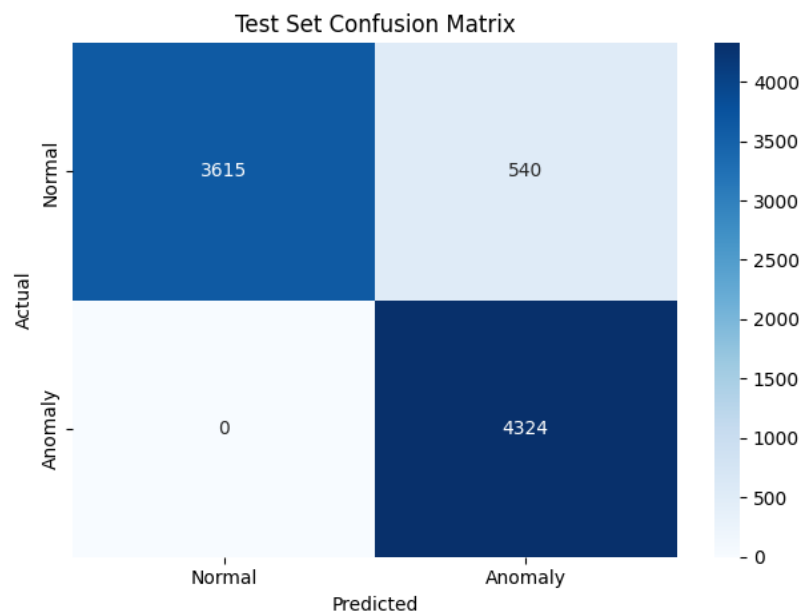
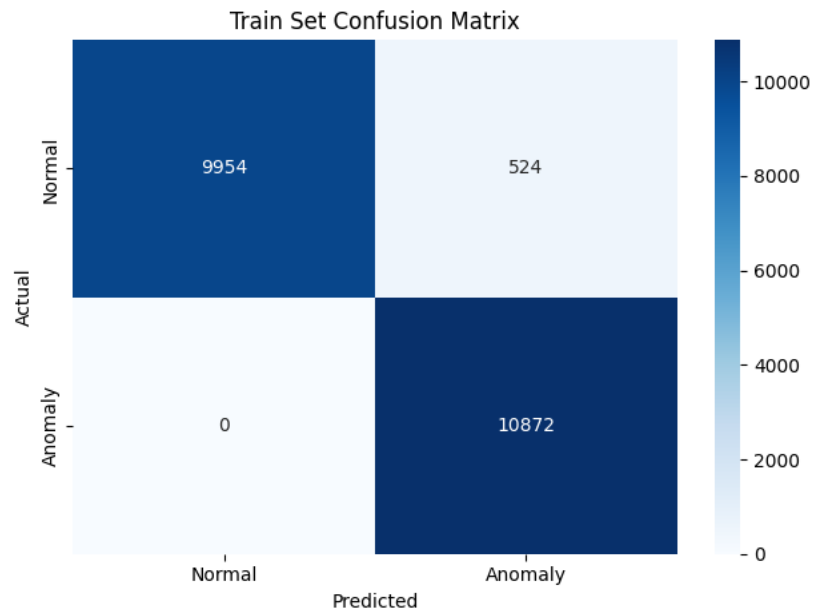
precision recall f1-score support

Normal (0)	1.00	0.95	0.97	10478
Anomaly (1)	0.95	1.00	0.98	10872
accuracy		0.98		21350
macro avg	0.98	0.97	0.98	21350
weighted avg	0.98	0.98	0.98	21350

Test set evaluation

precision recall f1-score support

Normal (0)	1.00	0.87	0.93	4155
Anomaly (1)	0.89	1.00	0.94	4324
accuracy		0.94		8479
macro avg	0.94	0.94	0.94	8479
weighted avg	0.94	0.94	0.94	8479



Train set evaluation: int8 model

	precision	recall	f1-score	support
Normal (0)	1.00	0.95	0.97	10478
Anomaly (1)	0.95	1.00	0.98	10872
accuracy		0.98		21350
macro avg	0.98	0.98	0.98	21350

weighted avg 0.98 0.98 0.98 21350

Test set evaluation: int8 model
precision recall f1-score support

Normal (0) 1.00 0.87 0.93 4155
Anomaly (1) 0.89 1.00 0.94 4324

accuracy 0.94 8479
macro avg 0.94 0.93 0.94 8479
weighted avg 0.94 0.94 0.94 8479

